BIO INSPIRED ALGORITHMS · 23DS7BSBIO

Cover Slide/Main Slide

Plant Disease Detection using Transfer-Learning CNN Optimized with Differential Evolution

INTELLIGENCE, NATURALLY.

Under the guidance of

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FIND THE BEST RECIPE

Introduction to Optimization Problems

01

Choose

Seven decisions define a training recipe: learning, regularization, architecture and augmentation.

02

Measure

Train on the training split. Measure macro-F1 on validation images only.

03

Improve

Search for a better recipe under a fixed evaluation budget.

VARIATION · EXPLORATION · SELECTION

What are Nature-Inspired Algorithms?

01

A population

Maintain several possible solutions instead of one guess.

02

Useful variation

Create trials by mutation and recombination.

03

Evidence-led selection

Keep a candidate when its measured fitness is no worse.

A MAP OF THE FIELD

Classification of Nature-Inspired Algorithms

01

Evolutionary

Genetic Algorithm

Differential Evolution

Evolution Strategies

02

Swarm

Particle Swarm

Ant Colony

Artificial Bee Colony

03

Other families

Simulated Annealing

Biogeography

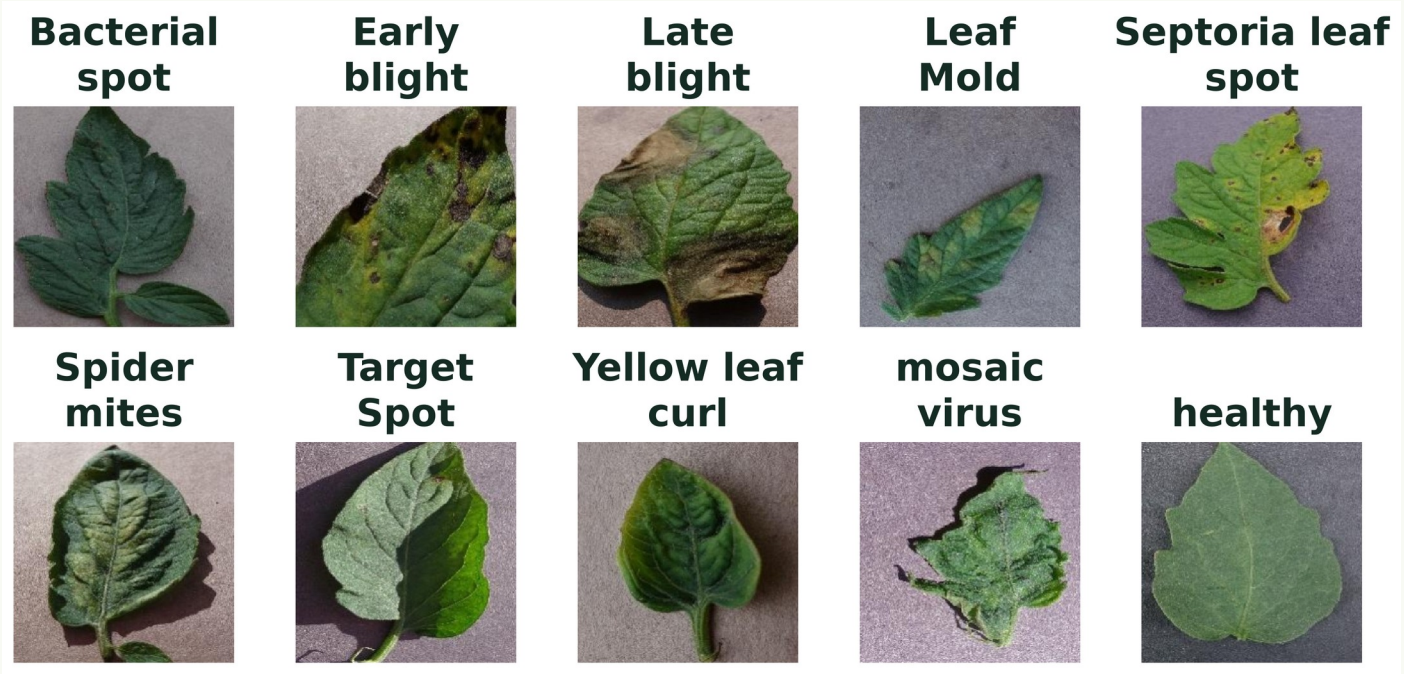
Teaching–Learning

TOMATO LEAF HEALTH

Problem Statement

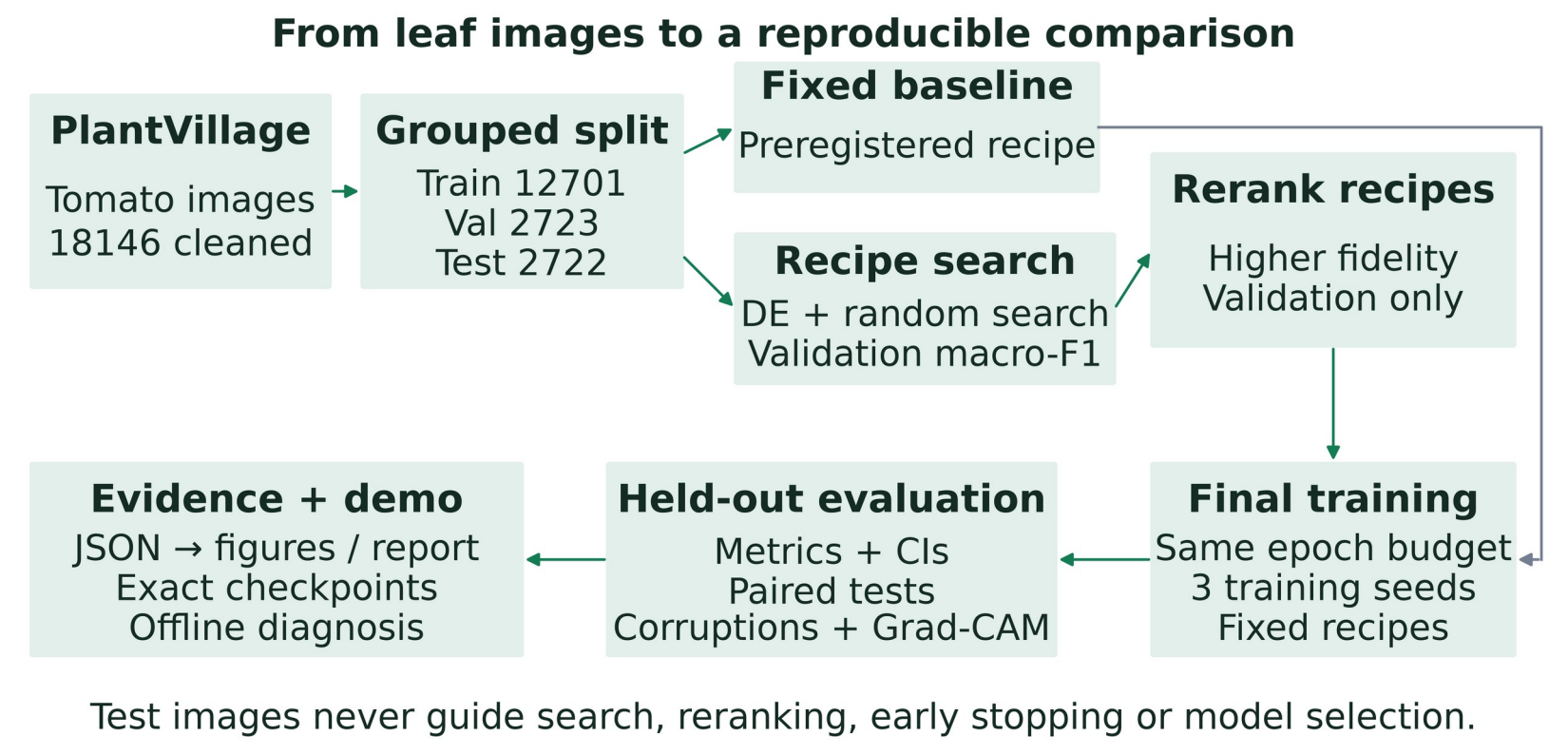
Evolve seven training decisions. Maximize validation macro-F1.

18,146 images · 10 classes · group-aware 70 / 15 / 15 split · 14 exact duplicates removed



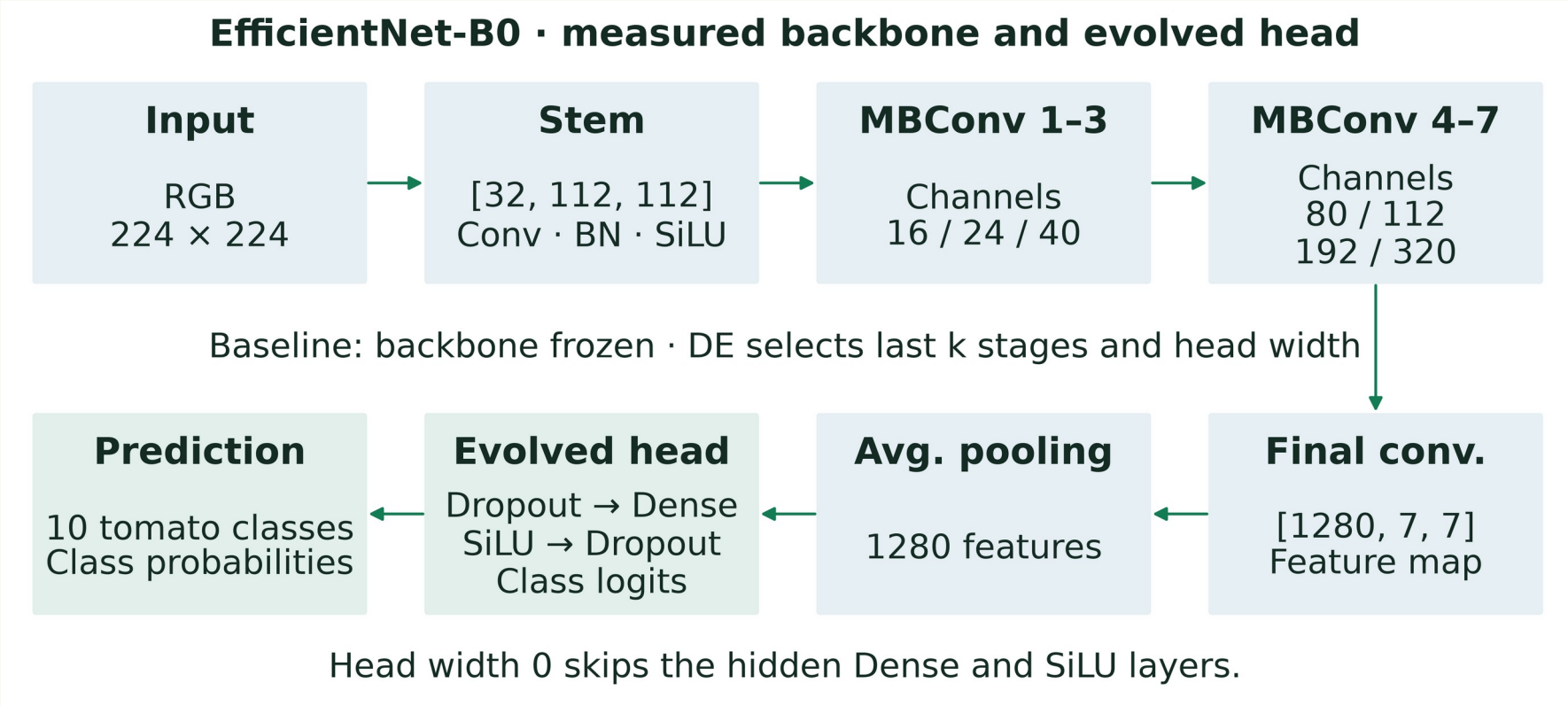
THE FULL SYSTEM

Implementation and Results (1/10)



A PRE-REGISTERED BASELINE

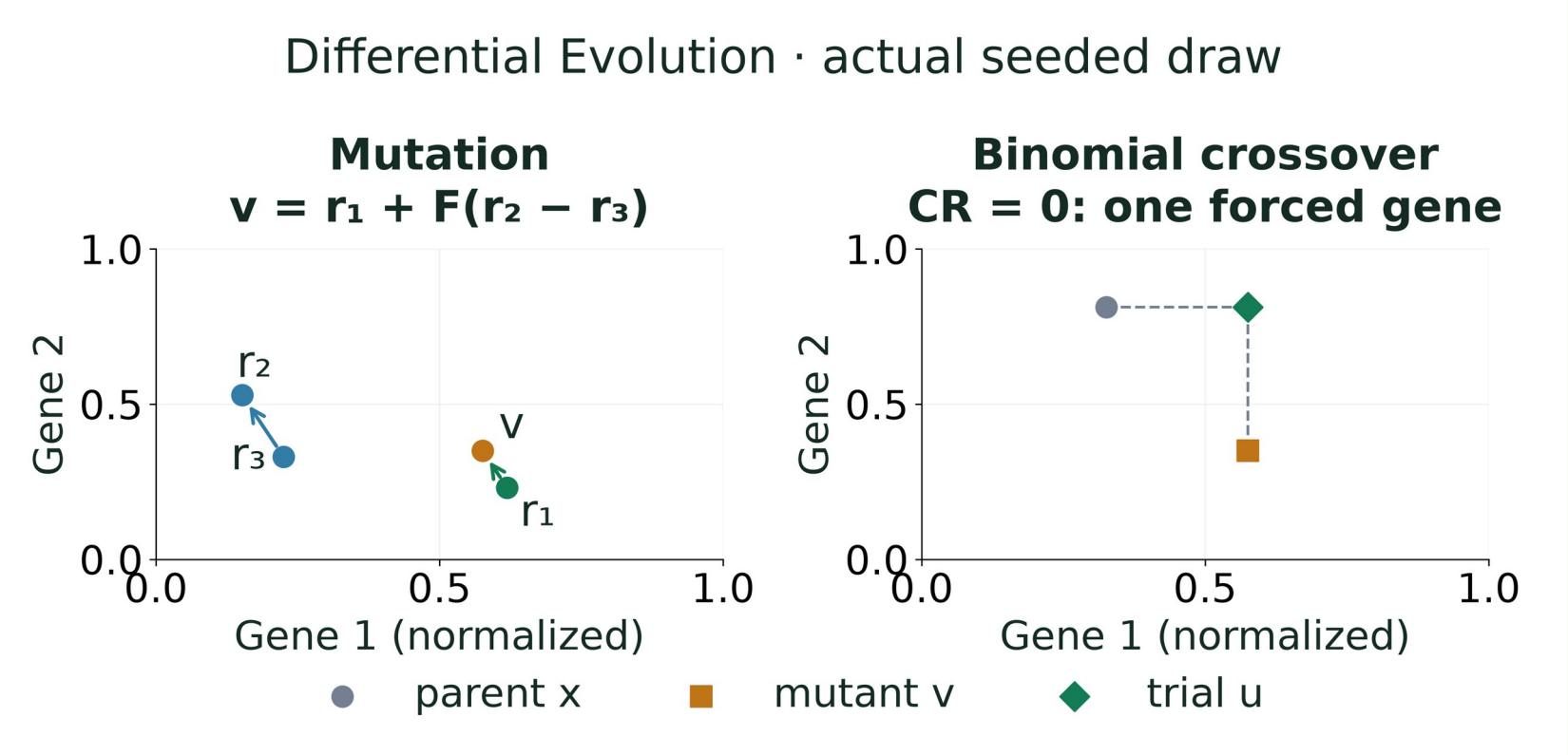
Implementation and Results (2/10)



The head and the number of unfrozen feature blocks are architecture genes.

DIFFERENTIAL EVOLUTION, FROM SCRATCH

Implementation and Results (3/10)



THE CHROMOSOME AND ITS SEARCH BUDGET

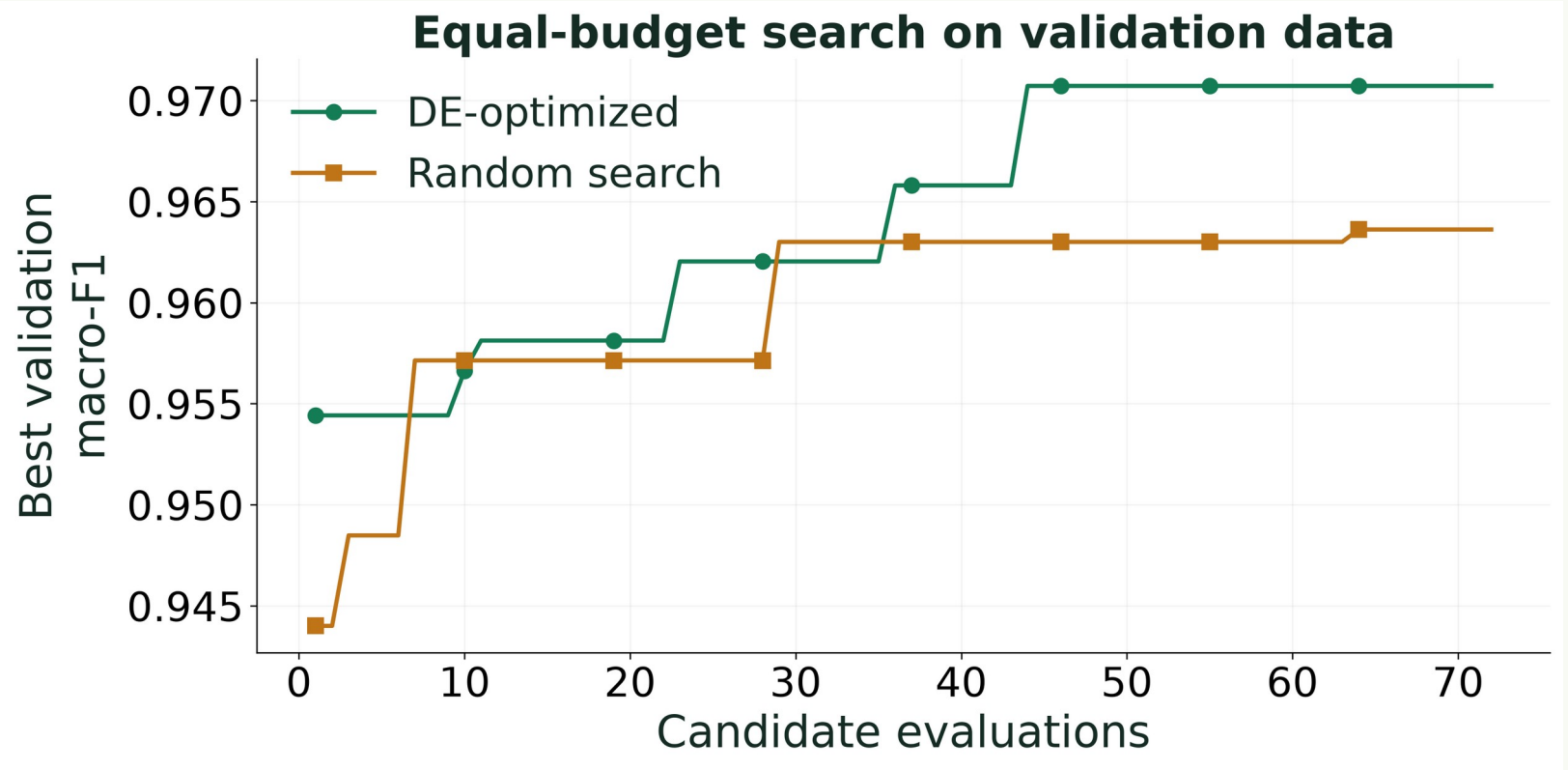
Implementation and Results (4/10)

GENE	DECODED CHOICE	SCALE
Learning rate	$10^{-4.5}$ to 10^{-2}	Logarithmic
Weight decay	10^{-6} to 10^{-2}	Logarithmic
Dropout	0.0 to 0.6	Linear
Head width	0, 128, 256, 512	Categorical
Unfrozen blocks	0 to 9	Integer
Label smoothing	0.0 to 0.2	Linear
Augmentation magnitude	0 to 15; 0 = no extra RandAugment	Integer

NP 8 · F 0.6 · CR 0.9 · 8 generations · 72 evaluations per method

SEARCH UNDER AN EQUAL BUDGET

Implementation and Results (5/10)

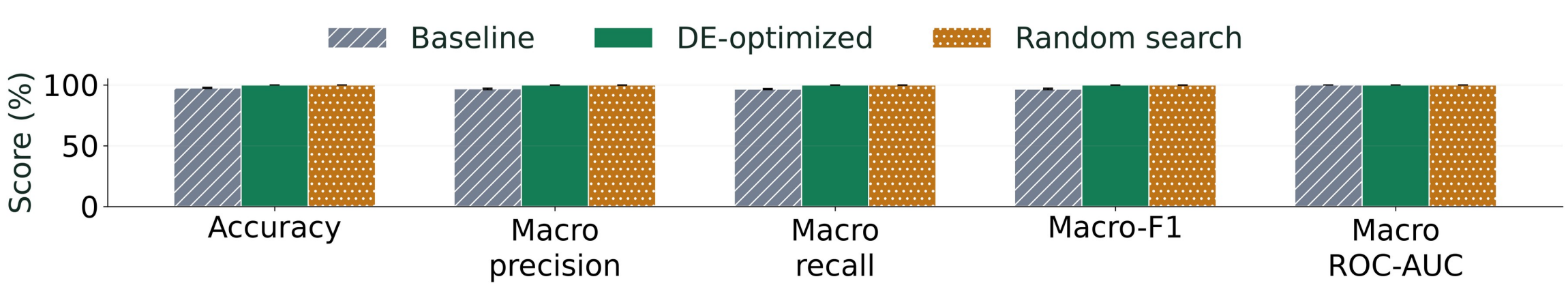


Proxy validation fitness; final test performance is evaluated separately.

BEFORE AND AFTER: MEASURED PERFORMANCE

Implementation and Results (6/10)

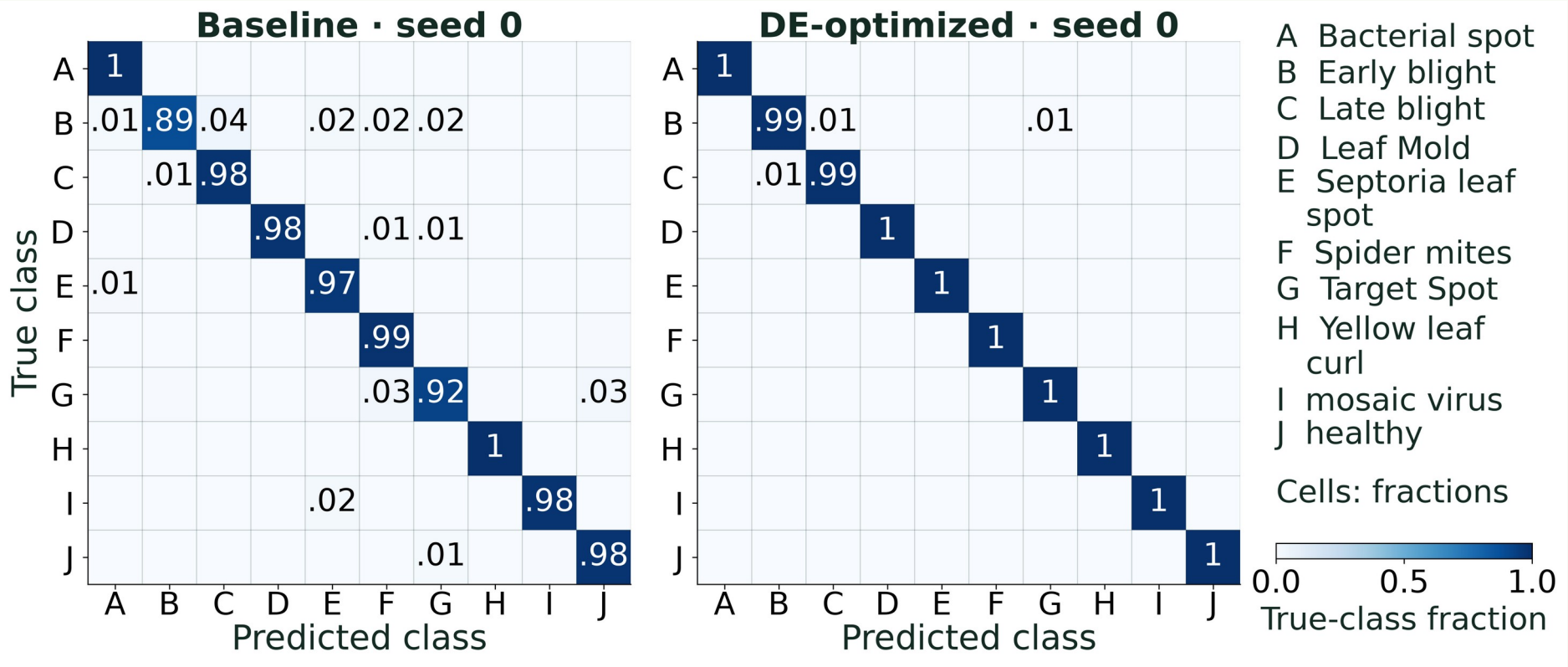
BASELINE ACCURACY	DE ACCURACY	BASELINE MACRO-F1	DE MACRO-F1
97.43%	99.85%	96.60%	99.80%
± 0.45 pp SD · 3 seeds	± 0.06 pp SD · 3 seeds	± 0.54 pp SD · 3 seeds	± 0.09 pp SD · 3 seeds



DE and RS tie at 99.85% mean accuracy · DE error reduction vs baseline: 94.29%

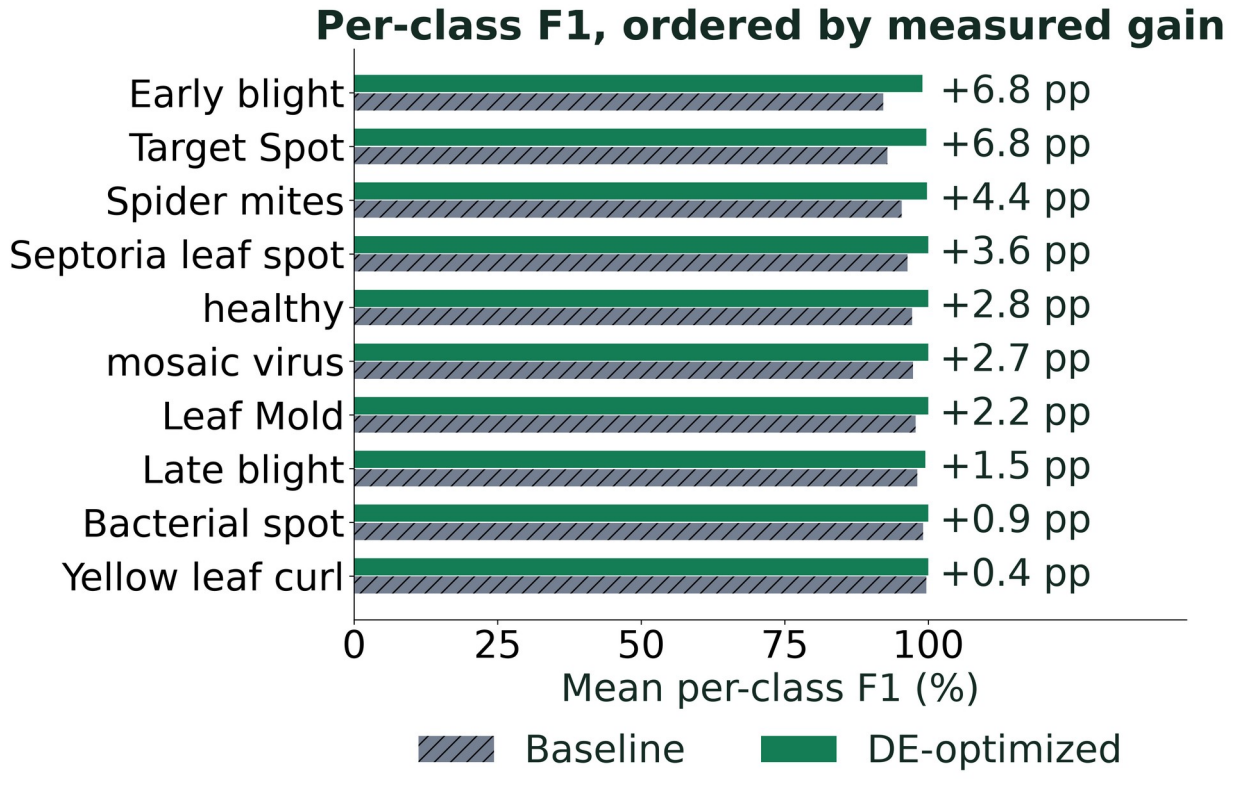
WHERE DO THE MODELS DISAGREE?

Implementation and Results (7/10)



EVERY CLASS COUNTS

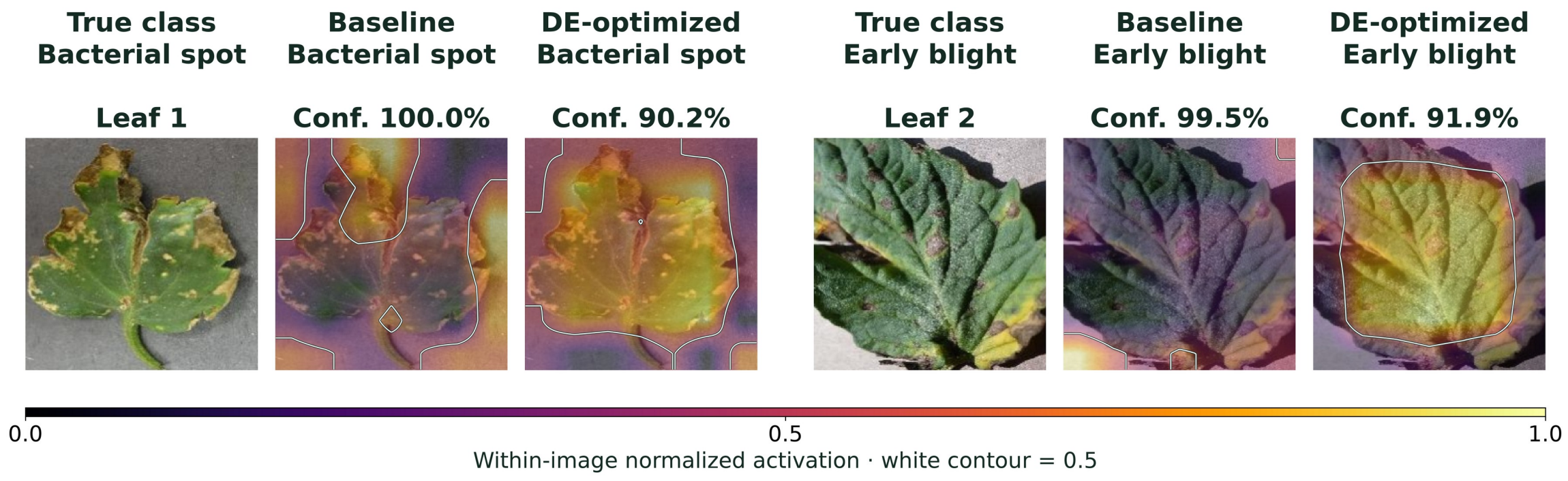
Implementation and Results (8/10)



Per-class F1 across all three seeds; classes ordered by measured gain.

WHERE THE MODELS LOOK

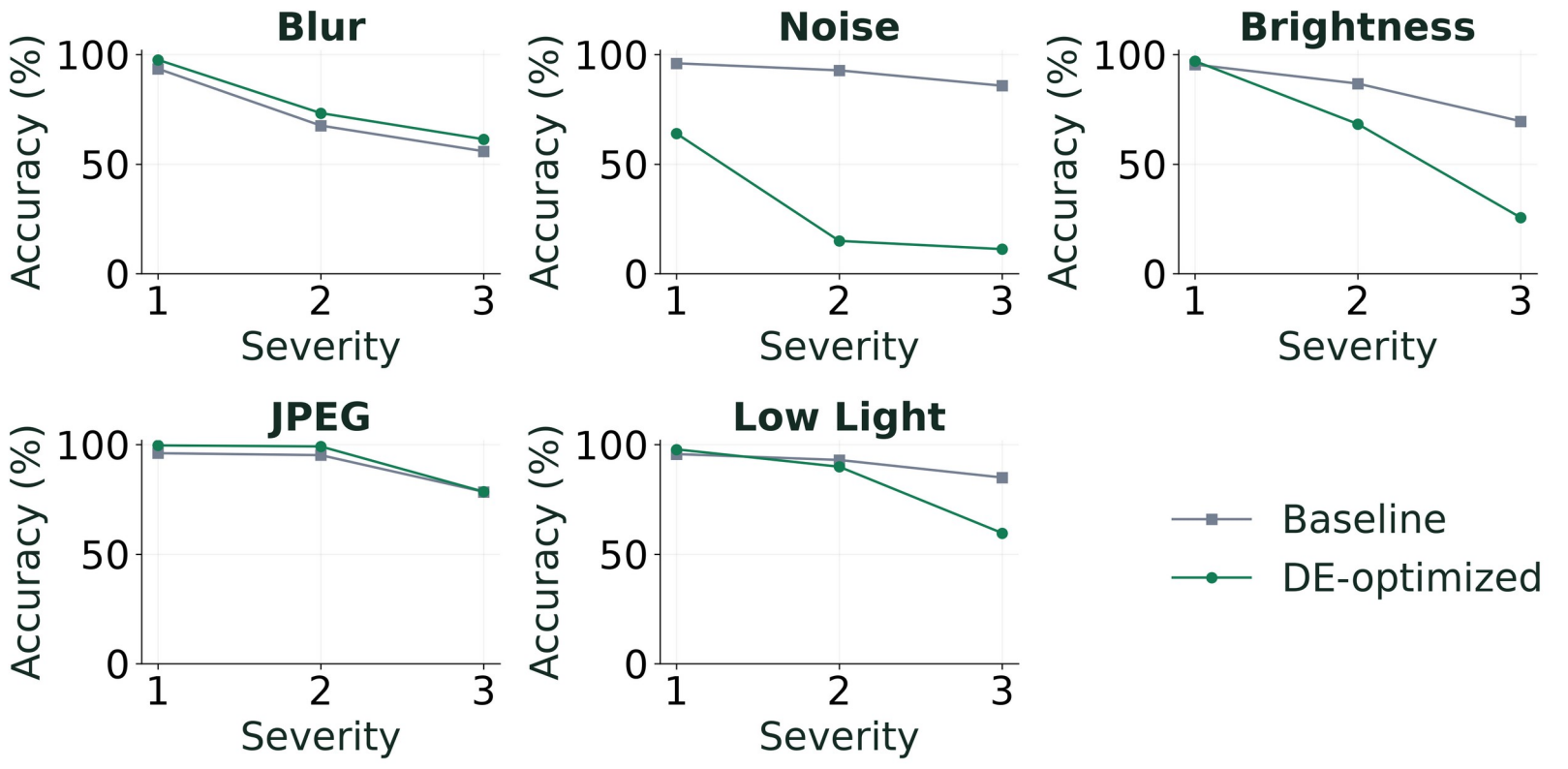
Implementation and Results (9/10)



Within-image normalized activation; attention is not a lesion mask or causal proof.

CLEAN ACCURACY IMPROVES; ROBUSTNESS DECLINES

Implementation and Results (10/10)



Mean corruption accuracy · seed 0, 15 conditions: baseline 85.77% · DE 69.19% — DE is worse.

PLANTWISE · INTELLIGENCE, NATURALLY

Live Demo

01

Diagnose

Compare exact baseline and DE seed-zero checkpoints on a sample leaf.

02

Explain

Inspect top-three probabilities and genuine Grad-CAM attention.

03

Evolve

Change the landscape and watch the population converge.

ACCURACY IS ONLY THE BEGINNING

Performance Metrics and Benchmarking (1/5)

01

Classification

Accuracy = correct / total

Precision = TP / (TP + FP)

Recall = TP / (TP + FN)

02

Macro-F1

F1 = 2PR / (P + R)

Macro-F1 = mean class F1

Every class counts equally.

03

Calibration

ECE · seed 0 · lower is better

Baseline 0.81% · DE 9.23%

RS 8.85%

DE has a larger calibration gap.

UNCERTAINTY AND PAIRED EVIDENCE

Performance Metrics and Benchmarking (2/5)

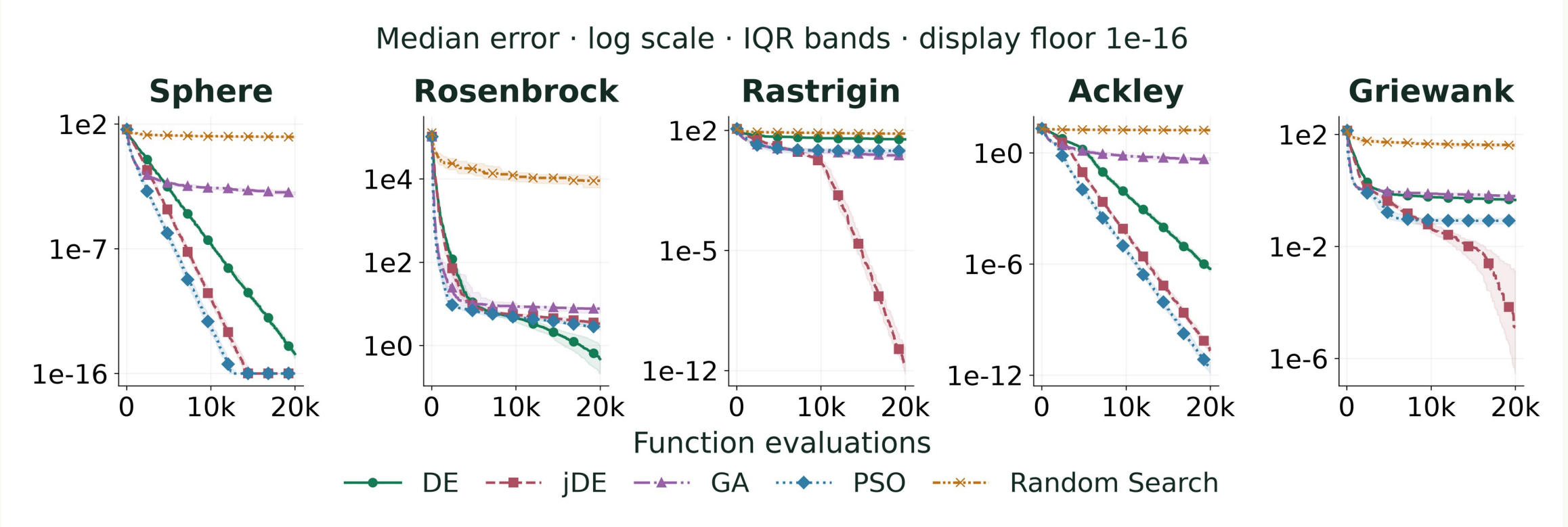
SEED-ZERO MODEL	ACCURACY · 95% CI	MACRO-F1 · 95% CI
Baseline	97.13% – 98.24%	96.15% – 97.70%
Random search	99.74% – 100.00%	99.69% – 100.00%
DE-optimized	99.63% – 99.96%	99.48% – 99.95%

Exact paired McNemar p = 9.048e-15

Baseline-only correct: 3 · DE-only correct: 60 · 1,000 bootstrap resamples

FIVE ALGORITHMS. EQUAL EVALUATIONS.

Performance Metrics and Benchmarking (3/5)



30 runs per algorithm/function · 20,000 evaluations per run · 10 dimensions

NO UNIVERSAL WINNER

Performance Metrics and Benchmarking (4/5)

FUNCTION	DE	jDE	GA	PSO	Random Search
Sphere	2.59e-15	1.27e-23	1.06e-03	2.59e-26	1.17e+01
Rosenbrock	4.72e-01	3.33e+00	7.66e+00	2.66e+00	8.99e+03
Rastrigin	3.00e+01	2.05e-12	3.30e+00	6.47e+00	6.44e+01
Ackley	5.46e-07	2.02e-11	4.47e-01	2.33e-12	1.66e+01
Griewank	4.57e-01	1.29e-05	6.22e-01	8.24e-02	4.10e+01

Median final objective · lower is better · 30 seeds per method and function

THE COST OF A BETTER RECIPE

Performance Metrics and Benchmarking (5/5)

MODEL	ALL 3 SEEDS · TRAIN	INFERENCE · MEAN ± SD	PARAMETERS
Baseline	30.5 min	4.71 ± 0.09 ms	4.34 M
Random search	66.1 min	4.80 ± 0.07 ms	4.67 M
DE-optimized	84.5 min	4.74 ± 0.05 ms	4.34 M

Search training: DE 45.9 min · RS 39.4 min

Batch-one latency on MPS · optimizer benchmark wall time: 105.3 s

WHAT THE EVIDENCE SUPPORTS

Advantages and Limitations

01

Advantages

Auditable operators

Reproducible data splits

Equal-budget control

Offline inference

02

Measured limitations

DE ties RS: 99.85% mean accuracy

DE has lower corruption accuracy

DE has a larger calibration gap

One crop/split; one search per method

FROM THE LAB TOWARD THE FIELD

Future Scope

01

Field evidence

Validate on real field images with an explicit class mapping.

02

Efficient intelligence

Optimize quality alongside latency and model size.

03

Adaptive learning

Explore SHADE, multi-fidelity search and expert feedback.

VERIFIED DOI SOURCES

References (1/2)

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Thank You / Questions

A healthier future.

One leaf at a time.

Plantwise · Intelligence, naturally.

github.com/Suraj-B12/BIO-INSPIRED-ALGO---AAT